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ANDREW K. SAYDJARI

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RESEARCH INTERESTS

I work at the interface of **data science** and **astrophysics**, developing new statistical tools to analyze large datasets. In terms of methods, I am intrigued by the low-SNR limit, **uncertainty quantification**, and blind signal separation. In terms of astrophysics, I strive to understand the spatial, kinematic, and chemical distribution of **interstellar dust**.

POSITIONS

Assistant Professor of Astronomy: New Mexico State University, Department of Astronomy — 2026 - present
NASA Hubble Fellow: Princeton University, Department of Astrophysical Sciences — 2024 - 2026

EDUCATION

Harvard University: PhD in Physics — 2018 - 2024
  Advisor: Douglas Finkbeiner
  Thesis: Statistical Models of the Spatial, Kinematic, and Chemical Complexity of Dust
Yale University: BSc/MSc in Chemistry, BSc in Mathematics — 2014 - 2018
  Thesis: Optimizing the Nickel-Catalyzed Carboxylation of Aryl Halides

SELECTED AWARDS & HONORS

International Astronomical Union (IAU) Thesis Prize: Division H (Galactic/ISM), Best PhD thesis	2025
NASA Hubble Fellowship	2024-2027
Eric R. Keto Prize (Harvard), Best PhD thesis in theoretical astrophysics	2024
Best Astrostatistics Student Paper Award (ASA/AIG)	2022
Bok Center Certificate of Distinction in Teaching (Harvard)	Fall 2021
NSF Graduate Research Fellowship (USA)	2018
Hertz Fellowship Finalist	2018, 2019
Howard Douglass Moore Prize (Yale), Chemistry's highest honor, awarded to a single graduating undergrad	2018
Barry Goldwater Scholar (USA)	2017
Phi Beta Kappa	2017
DAAD-RISE Fellowship (Yale/Germany), Research internship exchange	2016

PUBLICATIONS

I am an author on **50+ papers** that have **2274+** citations (h-index=19). This includes:
- **12+ papers** as **(co-)lead author** with 440+ citations
- **9+ papers** with **significant contributions** with 301+ citations
- **3+ papers** led by **students I (co-)advised** (denoted by * in my Publication List)

See my Publication List for details. My ORCID is 0000-0002-6561-9002.
Most of my papers can be found online on ADS, though citations outside astronomy are missing.

PROFESSIONAL ACTIVITIES & SERVICE

Collaborations, Committees, & Leadership:

Architect for SDSS-V, APOGEE pipeline development	Jan 2022 - present
Curriculum & Graduate Admissions Committees, Pizza Lunch Organizer (NMSU Astronomy)	Fall 2026 - present
Leadership Council, AI/ML Science Interest Group (NASA Cosmic Origin Program)	Fall 2025 – Summer 2026
Scientific Organizing Committee: SDSS-V 2026 Collaboration Meeting	Spring 2026
Computing Committee, Institute for Artificial Intelligence and Fundamental Interactions (IAIFI)	2022 - 2024
Faculty Search Committee, (1/2 elected) Student Representatives (Harvard Astronomy)	Jan - Mar 2023

Review:

External System Readiness Review (ESRR) Panel for the Fornax Initiative (NASA Cloud Compute) — Fall 2025
NSF Astronomy & Astrophysics Research Grant (AAG) Panel — Spring 2025
AAS Chambliss Poster Judge (AAS 240, AAS 241, AAS 245) — Summer 2022 - present
Journal Referee for Nature Astronomy, ApJ, AJ, A&A, and Journal of Open Source Software (JOSS) — 2023 - present

Conference and Seminar Organization:

Student Faculty Forum (StuFF) Co-organizer, Harvard Astronomy — 2022 - 2023

SUPERVISION & MENTORSHIP

I have (co-)supervised/mentored **9 students**, resulting in **3 student-led publications**:

Graduate:

5. Matt L. Wiemann (Sampson) (Astrophysics, Princeton) — Fall 2025 - Present
   Exploring noise curricula for ML optimizations (w/ Peter Melchior)
4. Christian Kragh Jespersen (Astrophysics, Princeton → UToronto Postdoc) — Spring 2025 - Present
   Constraining DIB carrier geometries via rotational spectra forward modeling
3. Rhys Seeburger (Astronomy, MPIA→ LJMU Postdoc) — Spring 2025 - Present
   Analyzing SB2s in APOGEE x Gaia (w/ Hans-Walter Rix)
2. Theo O'Neill (Astronomy, Harvard) — Fall 2024 - Present
   3D High Altitude Clouds (IVCs) and DESI Na Tomography (w/ Catherine Zucker)
1. Ana Sofía Uzsoy (Astronomy, Harvard) — Fall 2022 - Summer 2025
   Component Separation of Lyman Alpha Emitters in DESI (w/ Doug Finkbeiner)

Undergraduate:

4. Zack Steine (CS & Statistics, University of Toronto Scarborough → Veeva Systems) — Summer 2024 - Winter 2024
   SBI for DESI Stellar Parameters (w/ Josh Speagle)
3. Devisree Tallapaneni (Physics & Statistics, Cornell → OSU Grad) — Summer 2023 - Present
   Quantifying the Filamentary ISM: Statistical Reconstructions of Reality (w/ Eric Koch)
2. Stephanie Yoshida (Astronomy, Harvard → Milwaukee Brewers) — Fall 2023
   Kinetic Tomography of the Intermediate Velocity Arch (w/ Catherine Zucker)
1. Ken Michalek (Computer Science, Harvard Extension School → MIT Lincoln Lab) — 2020 - 2021
   Online Blind Deconvolution for Educational Astronomy (w/ Dominic Pesce & Allyson Bieryla)

Programs:

Polaris: Mentoring Harvard Physics Undergraduates (3 students) — 2021 - 2024

TEACHING

I care passionately about teaching and love ideating new ways of explaining difficult concepts. I emphasize the development of hands-on teaching methods, incorporating active learning through experiment and data-based exploration. I view creating an inclusive atmosphere, in which all students can comfortably learn, as a top priority.

Harvard University, Teaching Fellow — Fall 2021
   Solid State Physics, Lecture, Undergrad/Grad, 27 students, w/Prof. Julia Mundy
   Feedback: [Student Evaluations]

Yale University, Peer Tutor — 2015 - 2018
   Physical Chemistry, Lab, Undergrad, 30 students, w/Prof. Patrick Vaccaro
   Physical Chemistry II, Lecture, Undergrad, 30 students, w/Prof. Patrick Vaccaro
   Freshman Organic Chemistry II, Lecture, Undergrad, 100 students, w/Prof. Alanna Schepartz
   Sophomore Organic Chemistry I, Lecture, Undergrad, 120 students, w/Prof. Jonathan Ellman

SPLASH/SPROUT @ Yale, Middle School — 2015 - 2018
   Peeling Back the Layers of Solar Cells (30 students), Metal Mania: Simple Models of the Material World (4 students), Destressing Tensors (7 students), Abstract Algebra: Questions Teachers Didn't Answer (60, 75 students), Origins of Life: A Chemist's Perspective (16, 35 students)

SELECTED PRESENTATIONS

I have given **63+ talks (18+ invited talks/colloquia)**. See my Talk List for more details. Recent highlights include:

Invited Colloquia

Wesleyan Colloquium — February 2026
  Spectral Shadows of Structure at Scale: Chemokinematic Cartography of the Milky Way

UT Austin Colloquium — January 2026
  Spectral Shadows of Structure at Scale: Chemokinematic Cartography of the Milky Way

UW-Madison Colloquium — March 2025
  Mapping Milky Way Dust in n-Dimensions

NYU CCPP Seminar — November 2024
  The Spatial, Kinematic, and Chemical Complexity of Dust

Invited Talks

APOGEE's Galactic Odyssey — August 2026
  ISM Review Talk: Unveiling the Kinematics and Chemistry of Dust with APOGEE

SMI 2026: Statistical Methods in Imaging — June 2026
  Forward Modeling Linear Variable Filter Data: High Resolution PAH Maps from SPHEREx

AAS 247: PRIMA Special Session — January 2026
  PRIM(All): An All-Sky, Polarized, FIR Dust Map and Point-Source Catalog

Princeton Astrophysical Sciences Advisory Council Meeting — May 2025
  Galactic Cartography: Dust in 3D and Beyond

Roman GPS Community Workshop — February 2025
  Optimizing the Galactic Plane Survey Filter Selection

Galactic Science with the Nancy Grace Roman Space Telescope — June 2024
  The DECam Plane Survey as a Roman Galactic Plane Survey Pathfinder

Contributed Conference Talks

Dusty Universe: The 5th Pan-Dust Conference — November 2025
  Correlations between Extinction Features across Wavelength Scales:
  Realizing Diffuse Interstellar Bands as Chemical Tracers

JuliaCon Global — July 2025
  Building an End-to-End Spectral Reduction Pipeline for APOGEE

Interstellar Institute #7: Interstellar Physics Across Scales — July 2025
  Reconstructing the 3D+1V Velocity Field of the Milky Way from Dust Absorption

Sloan Digital Sky Survey V (SDSS-V) Collaboration Meeting — June 2025
  Mapping ISM Chemistry and Kinematics with APOGEE DIBs
  ApogeeReduction.jl: An APOGEE Reduction Pipeline for the AS5 Era

PRIMA and the Future of Far-Infrared Science — May 2025
  The Case for A PRIMA All-Sky Polarized Dust Map (and Point Source Catalog)

New Computational Methods in Milky Way Dynamics and Structure @ Ringberg — July 2024
  Bayesian Component Separation for Ground Based Spectra: Transforming Diffuse
  Interstellar Bands into Precision Kinematic Tracers

RAS Specialist Discussion: 1D ML — March 2023
  Measuring the 8621 Å Diffuse Interstellar Band in Gaia DR3 RVS Spectra

DECam at 10 Years Workshop — September 2022
  The DECam Plane Survey 2 (DECaPS2): More Sky, Less Bias, and Better Uncertainties

Seminars, Lunch Talks, & Journal Clubs

Columbia: Pizza Chalk Talk — February 2025
  The Spatial, Kinematic, and Chemical Complexity of Dust

IAS: Bahcall Lunch — November 2024
  The Highest Angular Resolution 3D Dust Map

OUTREACH & ENGAGEMENT

Public Science Writing

MathStatsBites: TheSequencer, CycleStarNet, SCMA8, NestedSampling — 2022-2023
LightSound Workshop, Soldering Solar Eclipse Sonification Instruments — Summer 2023
Cambridge Science Festival, MIT Museum Presentation Volunteer — Fall 2022
Latino Initiative Program, Instructor — Summer 2021- Summer 2023
Harvard Observation Project, Software Mentor — 2020-2021

GRANT SUPPORT & TELESCOPE TIME

7. Foundational Value-Added Data Products for the Roman Galactic Plane Survey — 2027 - 2030
   Co-Investigator (PI: Zucker), $1,200,000 — NASA (Roman Cycle 1 GI)
6. Highly Parallelized Grism Spectroscopy for Stellar and Extragalactic Astronomy — 2026 - 2029
   Co-Investigator (PI: Hogg), $98,000 — STScI (JWST Cycle 5 Archival)
5. Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey — 2025 - 2026
   Co-Investigator (PI: Schlafly), $35,000 — STScI (JWST Cycle 4 Director's Discretionary Time)
4. Topological Mapping of Super bubbles and ISM Structures with JWST — 2025 - 2027
   Co-Investigator (PI: O'Neill), $105,000 — STScI (JWST Cycle 4 Archival)
3. A Flexible Open-Source Framework for Rapid Stellar Classification in the Era of Roman — 2025 - 2027
   Co-Investigator (PI: Zucker), $388,000 — NASA (ROMAN24-ROSES)
2. A Next-Generation Crowded-Field Stellar Photometry Tool for Roman — 2025 - 2027
   Collaborator (PI: Smercina), $981,000 — NASA (ROMAN24-ROSES)
1. Inferring Kinematic and Chemical Maps of Galactic Dust — 2024 - 2027
   PI: Saydjari, $471,000 — STScI (NHFP Hubble)

PRESS

My research has been covered by 20+ news outlets worldwide, including the Wall Street Journal, CNN, and the Associated Press. Selected highlights:

IAU Thesis Prize: CFA, Princeton, IAU — June 2025
DECaPS2 Release: WSJ, Wired, AP, CNN, Register, Salon, Forbes, Space.com, AAS Nova — January 2023
Grad Student Highlight: Labroots — November 2022
Machine Learning & Interstellar Dust Clouds: Abstract: The Future of Science — December 2020

SELECTED RESEARCH SKILLS

Computational

I am a strong advocate of both open-source code and data, and I insist on public reproducibility of all plots in my work (see my Zenodo deposits accompanying my papers).

Developer: Julia (5 years, primary), Python (7 years), MATLAB (3 years) [Github]
  Developed pipelines and managed >100k core-h runs in both Julia and Python
  Managed daily simultaneous multi-instrument measurements in MATLAB
Public Packages: LowRankOps.jl, KryburyCompress.jl, CloudCovErr.jl, CloudClean.jl, EqWS.jl, crowdsource